Writing for Publication (WfP)

Week 1 – overall view of a scientific paper

Dr Adrian Rowland

Center for Language Education (CLE)

About me

Studied chemistry (MChem and PhD) at the University of Durham, UK

Joined SUSTech in July 2021

Teach both postgrad English courses + *Writing for Publication*

Language of this course

100% in English.

I know that many of you find it easier to read English than to listen to English.

→ I will put as much useful information as I can on these PPT slides for you to read.

I know that some of you might not have done much English speaking recently (or maybe ever). You might feel shy about speaking to me, or find it hard to speak to me. However, it is still helpful to me as part of my teaching to ask you questions from time to time.

→ Please do your best to answer my questions when you can, even if your answer is simple. I will always be grateful for your spoken contributions to this lesson.

→ I will try not to put you under individual pressure to answer when you can’t!

Well done for choosing this course

What percentage of their work time do you think the average professor spends writing, reading, and editing papers?

Academic scientists may spend up to half of their time writing, reading, and editing papers, and thus should be viewed to a substantial extent as professional authors, but fewer than 5% of them have ever received training in scientific writing.

David Lindsay, *Scientific Writing = Thinking in Words*, CSIRO Publishing, Melbourne, 2020

This course will help with writing papers, and it will also help with all other types of scientific writing: theses, lab reports, proposals, etc.

Course contents

Weeks 1-7 approx. The structure of a scientific paper, some relevant grammar, AI and other tools for supporting writing

Weeks 8-14 approx. Technical writing skills: paragraph construction, flow between sentences, concision, etc (will help with **all** writing, not just papers)

Weeks 15-16 approx. Summing up and preparing for the final exam

To some extent about papers, and to some extent about scientific English.

To sum up the course in two sentences:

Writing is easier to read when it contains the information the reader expects to see, in the position the reader expects to see it. So, most of this course is about what information to include and where to put it – in whole papers, in paragraphs, and in sentences.

How was this course designed?

Combination of theory and practice.

Theory is combined from:

- 5 books about scientific writing
- 3 books about style in English writing
- 1 writing course from *Nature*

Theory was refined using the following practice:

- I used the theory to help edit 20 scientific papers for SUSTech faculty and SUSTech postgraduate students, mostly in the fields of chemistry and computer science
- The course theory was reformed based on that experience
- What is taught in this course are ideas that I found useful when editing faculty and student writing
- Further updated every year based on progress of AI – course focuses mainly on areas where LLMs are less reliable (e.g. meaning, logic, emphasis) and less on areas where they are stronger (e.g. grammar)

A general scientific writing course

This course gives general ideas that apply, more or less, to most scientific writing.

You will need to interpret them a little for your own subject.

For instance, computer science papers are quite careless with tense and have a very distinctive introduction structure.

But for this course, I will teach conventional tense and a general approach to introductions, because that will be useful to the majority of students (and it will be *quite* useful to computer scientists too).

Thinking about any differences between what this course says and what your subject does may be instructive.

There is no required text

No required text – everything needed will be taught

If you want an alternative perspective on any of the topics we cover, you could try:

English for Writing Research Papers – Adrian Wallwork
https://ebookcentral.proquest.com/lib/sustechcn/detail.action?pq-origsite=primo&docID=4439459

The Scientist's Guide to Writing – Stephen B. Heard
https://ebookcentral.proquest.com/lib/sustechcn/reader.action?docID=5043174&ppg=133#

The Craft of Scientific Writing – Michael Alley
https://ebookcentral.proquest.com/lib/sustechcn/reader.action?docID=5327233

Structure of a typical lesson

Part 1: I’ll teach you something

Introduction of new topic
Examination of examples from scientific papers
Exercises
Discussion

Part 2: time for you to write

Work on your **mini-paper** (an ongoing writing task, I’ll explain it later) with advice from me

Aims for today

Admin

~~Course content and structure~~
Seating plan, communication, attendance, and assessment

Overall structure of a scientific paper

Please fill in your **name and student number** on the seating plan. Either:

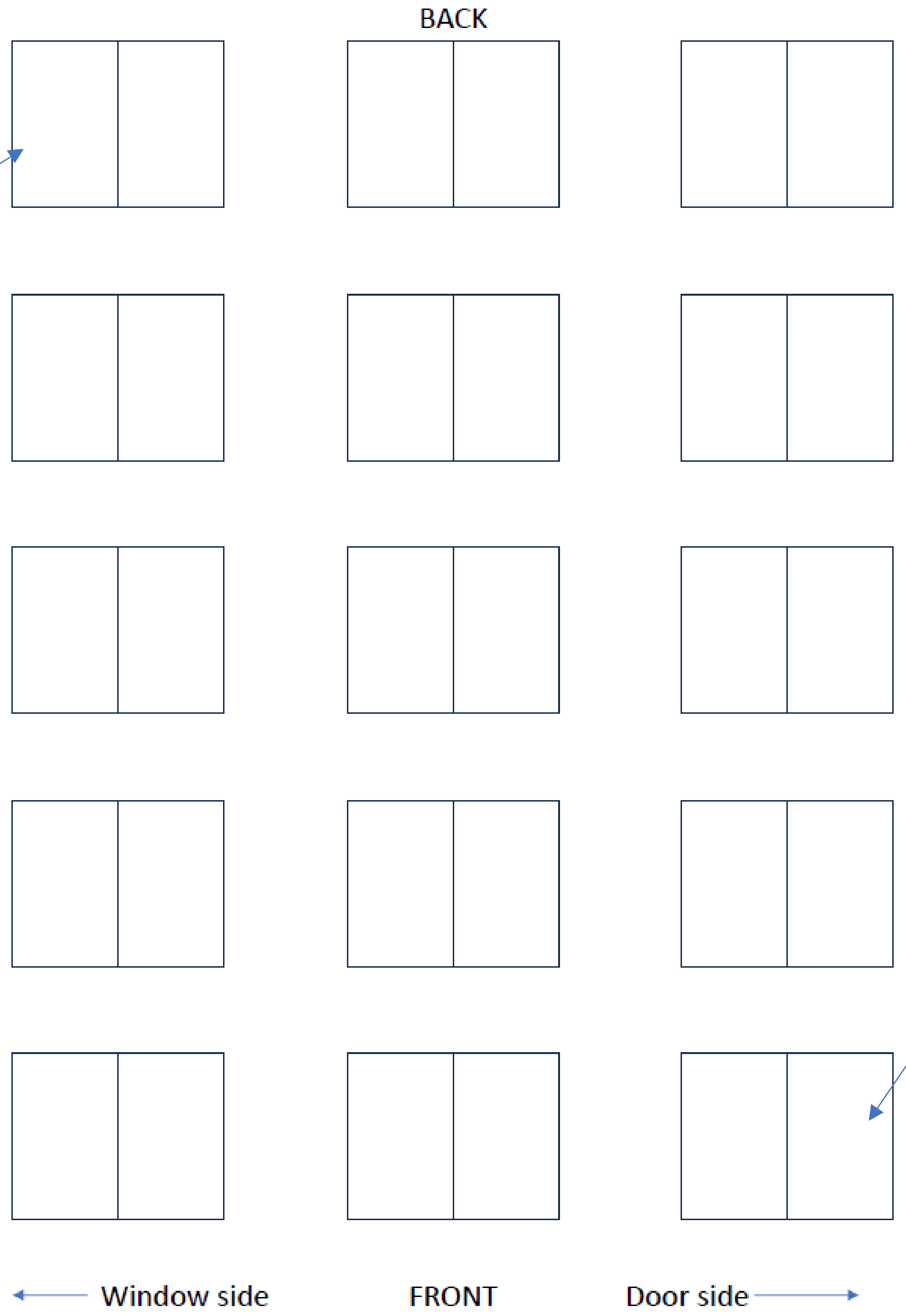
- An English name if you have one and wish to use it

- Or 拼音 if you don’t have an English name

- Please don’t just use 汉字 because I will be able to read (at most) 10% of them

We will fill out a seating plan like this **every lesson.**

Student sitting at back on window side is here



Student sitting at front on door side is here

The seating plan has two purposes:

1. I will use to it to (try to) learn (a few of) your names.

2. I will use it to calculate your engagement grade (so make sure you write your name in the place you are actually sitting).

   The plan needs to make sense to me, looking at the room from my point of view.

Please note: obviously I don’t mind where you sit, and you can sit in a different place every week if you want to. You do **not** have to sit in the same place every week (often students get confused and think that because there is a seating plan, they must stay in the same place every week).

Each week, wherever you are sitting that week, please record your position accurately on the seating plan for that week.

Communication – whole class

WeChat group

I will use it to give information and resources to the whole group

You should use it for any communication which might be relevant to the whole group

If you need to contact me individually

Questions, requests for help, office hours, etc

Either use email: adrian@sustech.edu.cn

Or add my WeChat from the class group

Both are fine

Communication – course material

Relevant material from class → Blackboard (doesn’t exist yet, should exist by next week)

When?

Materials will be placed on Blackboard after I have finished teaching the relevant lesson to all my classes.

Due to the unusual arrangement of holidays this semester, quite often some of my classes will be a week behind schedule. This means it may take up to a week for course materials to appear on Blackboard after your lesson.

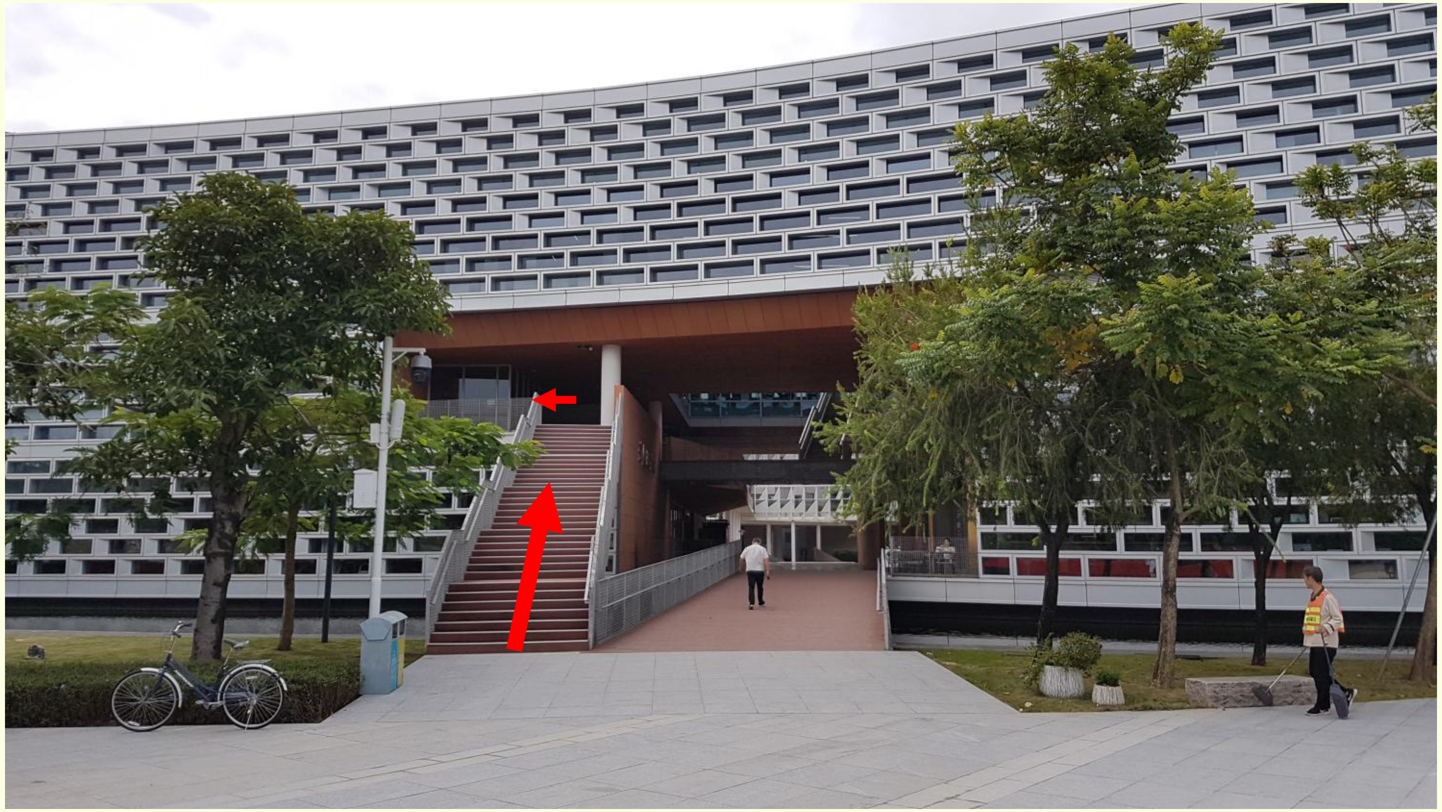
Office hours

My office hours are not yet fixed. I will tell you about them next week.

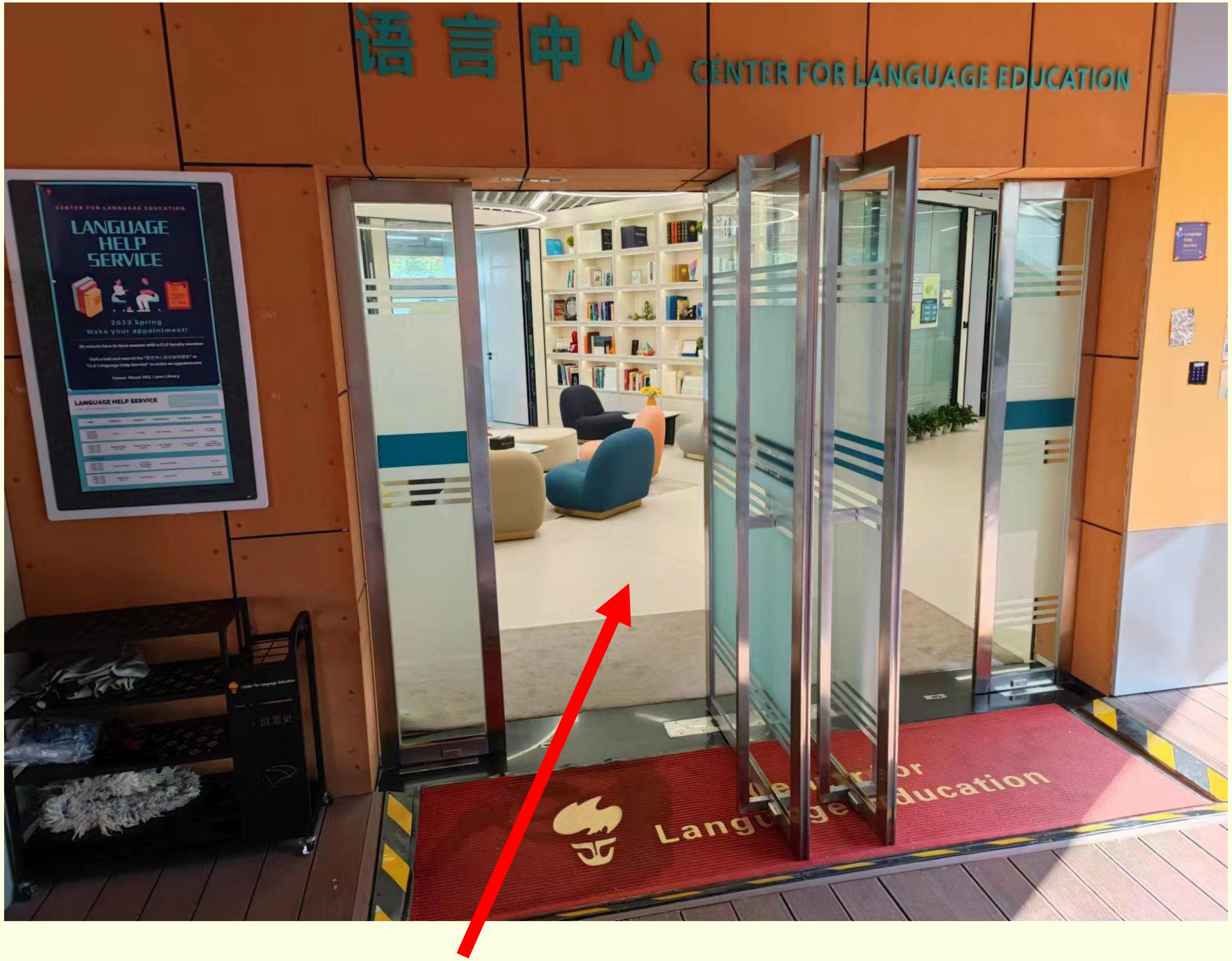
To find my desk in the CLE:

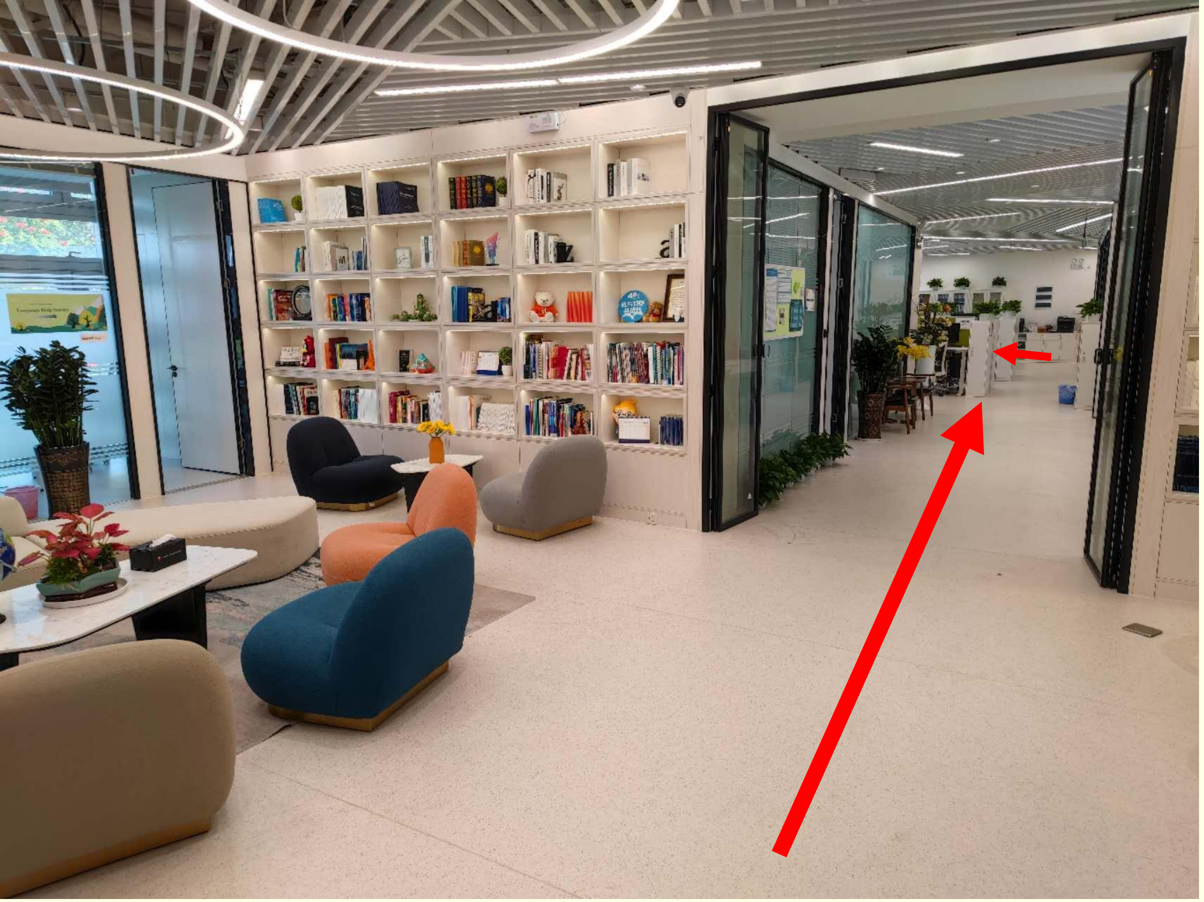
Go to the Lynn library

Go up these stairs and turn left at the top

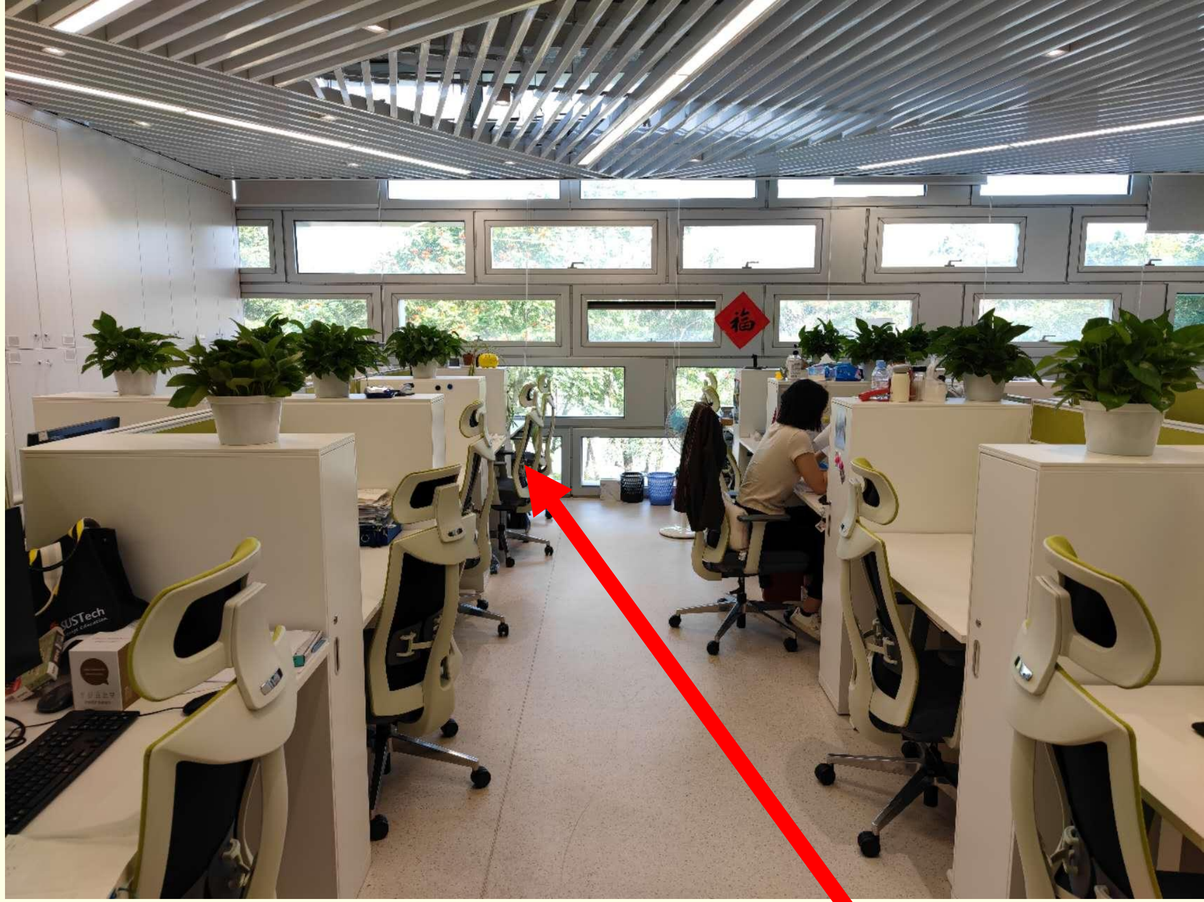


Enter the CLE





Turn right



I’m on the left of
the second aisle

Equipment for this class

To every lesson, you should bring:

- a pen
- a laptop/tablet

Assessment

Participation		10%
Engagement and performance		20%
Mini-paper	Submit in week 6	20%
Final exam	Week 17 (postgrad) Week 16 (undergrad)	50%

Assessment of participation (10%)

Standard CLE rules:

Full attendance = full participation marks

There’s a Center for Language Education (CLE) policy for this that I have to follow…

CLE attendance policy in detail

Arrive late (<= 20 minutes) = 0.25 absence

Arrive > 20 minutes late = 1 absence

Miss class = 1 absence

Includes after the break

1 absence	=	-2% to participation mark
2 absences	=	another -2% to participation mark (so -4% in total)
3 absences	=	another -2% to participation mark (so -6% in total)
4 absences	=	fail the course

CLE attendance policy in detail

Arrive late (<= 20 minutes) = 0.25 absence

Arrive > 20 minutes late = 1 absence

Miss class = 1 absence

Unofficially, I will ignore this part **as long as** there is no major problem with lateness.

If 2-3 people are a little late to class, I will ignore it. Having a few people arrive late does not disrupt the class and is not a big problem.

If more than 3 people are a little late, this is disruptive. In this circumstance, I will apply the policy for everyone who is late.

1 absence	=	-2% to participation mark
2 absences	=	another -2% to participation mark (so -4% in total)
3 absences	=	another -2% to participation mark (so -6% in total)
4 absences	=	fail the course

CLE attendance policy in detail

Arrive late (<= 20 minutes) = 0.25 absence

Arrive > 20 minutes late = 1 absence

Miss class = 1 absence

These parts will always be applied

1 absence	=	-2% to participation mark
2 absences	=	another -2% to participation mark (so -4% in total)
3 absences	=	another -2% to participation mark (so -6% in total)
4 absences	=	fail the course

Authorised absence (postgrads)

Absence does not affect your participation grade if it is authorised.

You get 2 free authorised absences – you can be absent twice with no problem. There's no need to ask my permission or to tell me about your absence. You don't need to do anything at all, your absence will be authorised, and your grade will not be affected.

For any further absences after the first 2 absences, the situation will be different.

For any further absences after the first 2, you need to <u>**email**</u> me to explain the reason for your absence, <u>**attaching a completed copy of the postgraduate school absence request form to your email**</u>. If you do this, your absence will (probably) be authorised. If you do not do this (email <u>**and**</u> form) your absence will be recorded as unauthorised and your grade will be reduced by 2% as described on the previous slide.

The form can be obtained from your department's teaching secretary.

Authorised absence (postgrads)

I will not accept **any** alternative to the postgraduate school absence form. I don’t want to see your medical records, your conference registration records, your train tickets, or a picture of your thermometer. Only the official postgraduate school absence form.

The deadline for sending absence emails to me is the Sunday following the lesson.

I will not accept late absence emails unless it was impossible for you to email me on time (e.g. you were unconscious in hospital).

Authorised absence (undergrads)

Absence does not affect your participation grade if it is authorised.

You get 1 free absence – you can be absent once with no problem, no need to apply for absence, nothing.

For any further absences, you need to apply for absence, using the online system, before the lesson.

Engagement and performance (20%) – how does it work?

If you treat this lesson like a lesson...

- You pay attention to me

- You have the equipment you need to do the tasks (pen + laptop/tablet)

- You fully take part in all tasks and activities

... You will get a good score.

A good score will be around 17/20.

If you also contribute to the lesson by offering answers to my questions in whole class discussion (or by asking me questions), you will get additional points and score >17/20.

Engagement and performance (20%) – how does it work?

If you treat this lesson as though it is a large lecture, or a boring meeting; if, for example...

- You don't pay attention to me

- You spend a significant amount of time playing with your phone, laptop, etc

- You sleep or wear headphones

- You work on your research, homework for other classes, etc

- You don't bother taking part properly in the tasks and activities I give you

- You can't properly take part in the tasks I give you because you don't have the necessary equipment

- Your conduct distracts me from teaching or other students from learning

... You will get a poor score.

How will I know what your score is?

That’s what the seating plan is for.

Each lesson, I will make some notes about engagement (mostly these will be positive, recording people who contribute to whole-class discussion).

Your engagement grade will be calculated from the notes on the seating plans.

How will you know what your score is?

I (probably) won’t comment on your engagement during the lesson. How much you engage with the lesson is your choice and I am not going to argue with you about it.

But I will notice and record your engagement, and I will give you all an engagement grade update on Blackboard in (approximately) weeks 4, 8, and 12, so you can see how you are doing.

The engagement grade **should** be an easy way to boost your overall score – if you just pay attention and have a reasonable try at tasks, your engagement score will be good!

Mini-paper (20%)

The name of this task is a bit misleading. It’s not really a paper.

Rather, you will write something in the style of a scientific paper **introduction** during the first few weeks of the course, then edit it in weeks 8-15 as a way to practise the writing skills we study in the course.

More on this next week.

Final exam (50%)

A test on all the ideas we cover in class.

You'll have practised all these ideas writing/editing your mini-paper.

Detailed revision guide and sample questions will be published beforehand so you can see the style.

No trick questions!

Writing test – no listening involved. This is a writing course only.

Grading policy for the whole course (not for undergraduates)

等级	A+	A	A−	B+	B	B−	C+	C	C−	D+	D	F
课程绩点	4.0		3.7	3.3	3.0	2.7	2.3	2.0	1.7	1.3	1.0	0
百分参考	95-100		90-94	85-89	80-84	77-79	73-76	70-72	67-69	63-66	60-62	0-59

According to regulations, approximately:

No more than 33% of students can get A+/A/A- grades
No limit on other grades

Aims for today

Admin

~~Course content and structure~~
~~Communication, attendance, and assessment~~

Overall structure of a scientific paper

Our example paper

Today and next week, we're going to look at some ideas about paper structure.

We need an example paper so we can see these ideas in context.

This paper will be our example this week and next week.

Usually when you study a paper, you are trying to get information from it.

Today, we will study this paper from a different point of view, a structural point of view:

- What *sort of* information does it contain?
- Where?
- For what purpose?

Although this paper is not from your discipline, most scientific papers share the same fundamental structure, and we can find that fundamental structure in this paper.

Our example paper

Rest of today’s lesson:

30 mins to get some understanding of example paper → then I teach some ideas about paper structure → then we look for this structure in the example paper

Preparing to analyse the paper

For the next 30 minutes, we're going to go through the paper together and summarise each paragraph in a simple way so that we can all understand the paper enough to analyse its structure.

The language of the paper is hard, but the ideas are simple, so if we can summarise it all in simple language, we will all get enough understanding.

I could just give you a summary, but it will be easier to appreciate the summary if you've at least tried to read the paper.

Similarly, you could just ask Chat-GPT for a summary – but again, if you just do this, you won't really understand the paper very deeply!

So, for each paragraph, I will give you a few minutes to read it and (if you have time) attempt to write a very short summary in simple language. Then I'll give you my summary of the paragraph.

If you don't finish (or even start!) your summary before I display mine, it doesn't matter. All that matters is that you *think about* each paragraph a bit.

Let’s begin

Please read paragraph 1 and (if you have time before I show my summary) try to write a brief (1-2 sentence) summary in simple English to express its basic meaning.

If you use translation tools/AI to help you, make sure that they are helping you to understand the point, not helping you to avoid needing to understand the point. Make sure you are **thinking**.

P1

Hint for writing P1 summary: when we meet new people, what’s the most important way we judge them?

When we meet new people, we judge them primarily according to how “warm” they are.

I’ll send you a copy of this slide (and all the summary slides, as we go through them) so you have a record of the overall paper summary.

Please now briefly summarise, in simple English, paragraph 2.

P2

Hint: Define "warm". What simple adjectives would you use to describe a warm person?

We think someone is "warm" if we believe they will be friendly, helpful, and trustworthy towards us. Basically, "warm" = "nice".

P3

Hint: the central idea of this paper is that there is an association, a link, between nice (warm) behaviour and physical warmth. Where does the association between physical warmth and nice behaviour come from?

This is the hardest paragraph. Understanding this paragraph is key to understanding the paper.

The key part of this paragraph is its final sentence. We describe friendly, helpful, trustworthy people as “warm” because of our experiences in early life.

In early life (when we were babies and small children) the people who provided us with friendliness, helpfulness, etc, (our “caregivers”, e.g. our parents), would also have provided physical warmth when they held us. We therefore form an association between friendliness, helpfulness, etc and physical warmth, creating the concept of interpersonal warmth (also referred to later in the paper as psychological warmth); this is why we describe nice people and nice behaviour as “warm”.

Basically, because our parents (i) held us, making us feel warm and (ii) were nice to us, we strongly associate being physically warm with nice behaviour.

P4

Hint: in these cruel experiments conducted by Harry Harlow, why did baby monkeys separated from their mother prefer a cloth fake mother to a wire fake mother? And what do you think is the purpose of this paragraph?

Baby monkeys separated from their mother preferred to stay close to a fake mother covered in cloth, rather than a fake mother made from wire, because the cloth mother was heated. Presumably, they expect warm behaviour from the physically warm mother.

(Though it doesn't actually say so, the point of this paragraph is to offer evidence to support the association between physical and psychological warmth introduced in P3.)

P5

Infants have a very strong need to be in physical contact with their caregiver.

P6 and 7

Hint: the first sentence of P6 is just repeating paragraph 3: because our parents held us (making us feel physically warm) and were nice to us, we have a strong association between nice behaviour and feeling physically warm.

How does the rest of the paragraph + P7 support a link between physical and psychological warmth? Extra hint: the insular cortex is a part of the brain.

The rest of P6 + P7:

Brain studies provide evidence for this hypothesis. They show that the same area of the brain (the insular cortex) processes both physical warmth information and interpersonal/psychological warmth information.

P8

What is the hypothesis, in simple language?

"For these theoretical and empirical reasons, we hypothesized that mere tactile experiences of physical warmth should activate concepts or feelings of interpersonal warmth.

→ Because of all this evidence of a link between physical and psychological warmth, touching a warm object should activate people's feelings of interpersonal warmth (I think this means it should prepare them to behave in a warm way).

Moreover, this temporarily increased activation of interpersonal warmth concepts should then influence, in an unintentional manner, judgments of and behavior toward other people without one being aware of this influence."

→ This will make people behave more nicely, and they won't be aware their behaviour has been affected.

P8-13

Experiments, results, discussion.

There were two experiments. Here is the simpler one:

Some students were given either a warm or a cold object to hold. Then they were given the choice between receiving a gift themselves or giving a gift to a friend. Students who held a warm object were more likely to give the gift to a friend; students who held the cold object were more likely to choose a gift for themselves.

In other words, students who had touched a warm object behaved more generously, i.e. more nicely.

P14

Start of P14:

“In summary, experiences of physical temperature per se affect one’s impressions of and prosocial behavior toward other people, without one’s awareness of such influences.”

I’m going to rewrite this slightly to make it easier for us to work with...

“In summary, experiences of physical warmth improve one’s impressions of and positive social behavior toward other people, without one’s awareness of such influences.”

What’s a simple way of expressing this sentence?

Touching a warm object makes people behave in a nicer way, and they are not aware that their behaviour has been influenced.

Overall summary of paper

Paragraph	Summary
1-2	Being warm means being nice.
3	We associate nice behaviour with warmth because the people who cared for us when we were very young were (i) nice to us and (ii) often held us (a physically warm experience).
4-5	Supporting evidence for P3 from studies on monkeys and children.
6-7	Supporting evidence for P3 from studies on the brain, which show that the same part of the brain processes emotions relating to psychological warmth and sensations relating to physical warmth.
8	Hypothesis: because there is a strong link in our minds between physical and psychological warmth, touching a warm object should make people behave in a nicer way without them being aware of it.
8-13	Experiments, results, discussion. In these experiments, people have better impressions of each other and are more generous when they have touched a warm object than when they have touched a cold object. One experiment also tested whether or not people were aware that their behaviour had been affected; they were not aware.
14	Summary: physical warmth makes people behave in a nicer way and they are not aware that their behaviour has been affected.

Hopefully you can see that, although the language of the paper is quite hard, the ideas and arguments of the paper are more straightforward.

A question

If we want people to understand our research and think it is important, our papers must make clear (i) what we have achieved and (ii) why it matters.

The paper we have just read achieved both these things pretty well.

How did it achieve these things?

How can we structure a paper to help make it clear what we achieved and why it matters?

Overall structure of a scientific paper?

What is the overall structure of a typical paper?

Scientific papers are often described as having an **IMRaD** structure:

(Abstract – summary)
Introduction – the background information needed to understand the research
Method – what you did
Results – what happened
and
Discussion – what it means

(In reality, the exact structure varies between different journals, and these section titles may or may not be used.)

How can we structure a paper to make it clear what we achieved and why it matters?

Our real question for today...

Most papers are written in the IMRaD model (or something similar).

Most of the time, it is not possible to adopt a wildly different structure – we have to follow IMRaD or something close to it.

So, our real question today is actually:

Working within the IMRaD model, how can we structure the whole paper to make it clear **what we've achieved** and **why it matters**?

The answer is: tell a story

Before I wrote my PhD thesis, my supervisor said to me “Remember, you are telling a story.”

Tell a story

If you read a book about scientific writing, it will probably say that a paper tells a story.

The Craft of Scientific Writing – Michael Alley

“Story”+ “stories” = 26

The Scientist’s Guide to Writing – Stephen B. Heard

“Story”+ “stories” = 77

Writing Science – Joshua Schimel

“Story”+ “stories” + “storyteller” = 20 in less than two pages, and then I stopped counting

Tell a story

Prof. Zheng in the chemistry department said to me “When my students are writing a paper, I tell them that they must write a story.”

(Or something like that – I can’t remember his exact words.)

What does “tell a story” actually mean?

It means your paper should have the basic structure of a story.

Let’s find out what that structure is.

Read the two stories on the handout provided.

Decide which story is better.

Stories

Story 2 is better:

Initial situation is explained → the reader is made curious → events occur → the reader's curiosity is satisfied

What's a simple structure that gives this effect?

Analysis of the story structure

Stories have an OCAR structure:

Often the question is **implied** – it's not a literal question with a "?", it's a statement that creates curiosity.

Opening – the background we need in order to understand the **challenge**

Challenge – the question or problem that the story/paper seeks to answer or solve

Action – how we get from the **challenge** to the **resolution** (the majority of the story)

Resolution – the answer to the question or the solution to the problem

Decide which sentences of story 2 are the opening, the challenge, the action, and the resolution.

The basic structure of a story

OCAR

There are multiple reasonable answers. Some suggestions:

Maybe the challenge is “why wasn’t the baker getting enough butter?” or “how was the farmer cheating the baker?”

O = 1, C = 2, A = 3-7, R = 8-10

Maybe the challenge is “what was the result of the court case?”

O = 1-2, C = 3, A = 4-9, R = 10

Notice that in both cases, the challenge is **implied** – the story doesn’t contain a literal question with a “?”.

How does OCAR fit in with IMRaD?

Opening	Background	**Introduction**
Challenge	What you wanted to find out – the research question(s)	
Action	What you did	**Method**
	What you saw	**Results**
	What it meant	**Discussion**
Resolution	The final answer(s) to the question(s)	

OCAR does not really present any new ideas about the contents of a paper – it's just a perspective that emphasises the importance of the Introduction and Discussion sections.

OCAR does not have much to say about the **action**.

The rest of this lesson

We’ll start to find the O, C, and R of OCAR in the Warmth paper, and thereby start to notice the paper’s story structure.

This is a two-part lesson: we will finish applying OCAR to the Warmth paper next week.

You might not really appreciate the point today – hopefully you will by the end of next week!

Starting to find the OCAR structure in the Warmth paper

1. Where is the challenge? Find the relevant paragraph. (The challenge is written as a statement, not as a question.)

2. When we read the challenge, we can guess that the paper is going to answer two simple questions by testing the hypothesis. What are these two simple questions?

3. Where is the resolution?

You will find it easier to answer these questions by looking at the summary table instead of the actual paper. You can find this table in our class 微信 group.

Overall summary of paper

Paragraph	Summary
1-2	Being warm means being nice.
3	We associate nice behaviour with warmth because the people who cared for us when we were very young were (i) nice to us and (ii) often held us (a physically warm experience).
4-5	Supporting evidence for P3 from studies on monkeys and children.
6-7	Supporting evidence for P3 from studies on the brain, which show that the same part of the brain processes emotions relating to psychological warmth and sensations relating to physical warmth.
8	Hypothesis: because there is a strong link in our minds between physical and psychological warmth, touching a warm object should make people behave in a nicer way without them being aware of it.
8-13	Experiments, results, discussion. In these experiments, people have better impressions of each other and are more generous when they have touched a warm object than when they have touched a cold object. One experiment also tested whether or not people were aware that their behaviour had been affected; they were not aware.
14	Summary: physical warmth makes people behave in a nicer way and they are not aware that their behaviour has been affected.

Hopefully you can see that, although the language of the paper is quite hard, the ideas and arguments of the paper are more straightforward.

1. Where is the challenge?

Start of P8 (the hypothesis of the paper).

2. Express the challenge as simple questions

This was our simple summary of the hypothesis:

Because of all this evidence of a link between physical and psychological warmth, touching a warm object should activate people's feelings of interpersonal warmth.

This will make people behave more nicely, and they won't be aware their behaviour has been affected.

When we read this hypothesis, we can guess that the paper is going to answer two simple questions. What are they?

When people touch a warm object...?

...?

When people touch a warm object, will they behave more nicely?

Will people realise their behaviour has been affected?

3. Where is/are the resolution(s)?

Start of P14: “In summary, experiences of physical temperature per se affect one’s impressions of and prosocial behavior toward other people, without one’s awareness of such influences.”

This was our simple version of this text:

“Touching a warm object makes people behave in a nicer way, and they are not aware that their behaviour has been influenced.”

Let’s use it to answer the questions of the **challenge**.

Will touching a warm object make people behave more nicely? **Yes**

Will they be aware that their behaviour has been affected? **No**

This slide shows (part of) the reason this paper is so successful (published in *Science*, > 600 citations): because it has a clear OCAR C and R, it is clear (i) what questions were asked and (ii) what the answers were. In other words, it is clear what this paper achieved.
(Though you could certainly argue they could have made it even clearer by using less complicated language!)

The resolution could actually be more clearly written

We’re not going to concentrate on this now, but if you take time later to review the paper in detail, you’ll find that the challenge and the resolution don’t match as well as they could.

The challenge is about **warmth**.

But the resolution is vaguer, just about **temperature** (warm and cold).

[When we looked at paragraph 14 earlier, you might remember that it said

“In summary, experiences of physical temperature per se affect one’s impressions of and prosocial behavior toward other people, without one’s awareness of such influences.”

I.e. it was about temperature. I rewrote it to be about warmth to make it a bit easier for us to work with:

“In summary, experiences of physical warmth improve one’s impressions of and positive social behavior toward other people, without one’s awareness of such influences.”]

How can we make it clear **what we've achieved** and **why it matters**? Summing up:

- To make sure the reader understands what the author has achieved and why it matters, a paper (or thesis) should tell a story
- A story has an OCAR structure
- Papers usually have some variation on an IMRaD structure, which matches up to OCAR
- The introduction of a paper contains the **opening** and the **challenge**
- The **resolution** comes near the end of the paper and should relate clearly to the **challenge**
- The **challenge** and **resolution** highlight **what a paper has achieved**

We are less than halfway through examining OCAR; all we have really said today is that a paper has a sort of question-and-answer structure. Next week, we will look in more detail at the **opening** and the end of the paper and their importance in showing the reader **why the paper matters**.

OCAR is also helpful when reading

Papers are often a confusing mass of complicated details.

Finding the challenge and resolution, and expressing both of them in simple language, can be helpful for giving you the right perspective to read and understand a paper.

If you cannot locate and clearly express the challenge(s) and resolution(s), this may indicate the paper is not very well written and that it is not your fault you are finding it hard to read!

Homework – voluntary

I’ve introduced OCAR and helped you to find (part of) it in a paper not from your discipline.

You might have found that quite hard.

You should find this exercise easier with a paper from your own subject, so examine a paper in your discipline. If you pick a well-written paper, one you found relatively easy to read, an OCAR-like structure should be obvious.

Can you clearly and easily identify the **challenge(s)**? Exactly what the challenge looks like will depend on your discipline. It could be a research question (or questions), a hypothesis, or a statement of a problem that needs to be solved, or a statement about something we don’t know…

Can you clearly and easily state the **resolution(s)**, the answer(s) or solution(s) to the challenge(s)?

Extension: look at another paper, one you found hard to read. Can you find a clear challenge and resolution now?